NIRIKA LAMICHHANE

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SUMMARY

Computer engineering student focused on machine learning, deep learning, and language AI. Developed NLP systems for Devanagari Nepali, including target-aspect based sentiment analysis. Interested in research at the intersection of AI and healthcare.

PROFILES

https://www.linkedin.com/in/nirika-lamichhane/
https://github.com/Nirika-Lamichhane
https://medium.com/@nirikalamichhane12

SKILLS

- Python and Django development
- Backend development
- API development
- API integration
- PostgreSQL database management
- Git and GitHub experience
- Agile methodologies
- Data analysis

EXPERIENCE

BACKEND DEVELOPER INTERN, 05/2026 - 06/2026

Akin Pte Ltd, Kathmandu

- Collaborated with frontend and product teams to define and clarify feature requirements, ensuring alignment with user needs.
- Wrote unit tests for backend modules to support code quality.
- Debugged code issues in ongoing projects, tracing root causes to enhance system stability.
- Maintained documentation of changes to backend infrastructure throughout development cycles, facilitating knowledge sharing and future reference.

JOINTSECRETARY AT TENSOR CLUB, 06/2023 - 06/2026

IOE, Thapathali Campus

- Coordinated with faculty and student teams to plan and execute AI and Data Science workshops, fostering enhanced learning and collaboration.
- Organized meetings around individual schedules, achieving high attendance and active participation across teams.
- Conducted interviews and selected new members, strengthening the team with committed and capable individuals.
- Managed club communications, sending personalized emails to keep members informed and engaged.

EDUCATION

IOE, Thapathali Campus, Tribhuvan University, Thapathali, Kathmandu, Nepal

Bachelor: Computer Engineering

Ranked in Top 1 of class

Related Coursework: Database System, Machine Learning, Data Science, NLP

PROJECTS

ASPECT-BASED SENTIMENT ANALYSIS FOR NEPALI LANGUAGE | PYTHON, CNN, BILSTM, APIS

GitHub: https://github.com/Nirika-Lamichhane/NLP-project

- Architected a hybrid CNN-BiLSTM model for aspect-based sentiment analysis targeting Nepali socio-political YouTube comments scraped via APIs.
- Engineered a robust NLP data pipeline to identify, transliterate, and standardize mixed-code and Romanized Nepali into pure Devanagari script prior to model training.
- Categorized complex sentiments across 5 custom, domain-specific aspects (Governance, Service, Corruption, Economy, and Policy) to extract nuanced public opinion insights.

MACHINE LEARNING ALGORITHMS FROM SCRATCH | PYTHON, MATHEMATICS

GitHub: https://github.com/Nirika-Lamichhane/LogisticRegression_FromScratch

- Implemented Principal Component Analysis (PCA) entirely from scratch, computing covariance matrices, eigenvectors, and eigenvalues to center and reduce the dimensionality of a generated 3D dataset.
- Developed a Logistic Regression model from the ground up without external ML libraries, training it efficiently on the PCA-reduced dataset.
- Demonstrated a deep theoretical understanding of core machine learning mathematics, feature scaling, and optimization techniques.

YATRA APP – TRAVEL RECOMMENDATION PLATFORM | DJANGO, POSTGRESQL, MACHINE LEARNING

GitHub: https://github.com/Nirika-Lamichhane/YatraApp

- Developed a scalable backend utilizing Django and PostgreSQL to support role-based registrations for regular users and local guides. Integrated custom machine learning algorithms to generate dynamic commute paths.
- Designed an extensible system architecture capable of seamless future integration with real-time traffic APIs.
- Developed a user-friendly frontend interface utilizing Kivy to deliver dynamic route suggestions for improved urban mobility under varying traffic conditions.

LEDGERLOGIC – BUSINESS FINANCE PLATFORM | C++, QT

GitHub: https://github.com/Nirika-Lamichhane/LedgerLogic

- Built a comprehensive financial management platform using C++ and the Qt framework for streamlining business accounting and balancing.
- Designed an interactive dashboard to automate the generation and tracking of bills, journals, and payments.
- Implemented robust ledger summary, trial balance, and balance sheet features to ensure accurate financial reporting and reconciliation for small businesses.

LEADERSHIP AND MEMBERSHIP

- Mentored school students in AI and ML workshops.
- Aspire Institute Leadership Program, Cohort 5 (2025).
- Sponsorship and Outreach Manager at OnCampus Hult Prize 2026.
- AI Fellow 2026, Fusemachine Nepal.
- Student Partner at Leapfrog, LSPP 2026.
- Research fellowship at NSSR (AI/ML track).

REVIEW PAPER

GRAPH NEURAL NETWORKS (GNN) REVIEW PAPER | RESEARCH & ACADEMIC WRITING

GitHub: https://github.com/Nirika-Lamichhane/GNN-review-paper

- Co-authored a comprehensive review paper analyzing the history, foundational frameworks, and evolutionary milestones of Graph Neural Networks, including their applications in sectors like finance.
- Synthesized complex academic literature and structural frameworks following insights gathered from the ANAIS 2025 conference.